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The cells of beetroot contain a red pigment. A student investigated the effect of temperature on the loss of red pigment from beetroot. He put discs cut from beetroot into tubes containing water. He maintained each tube at a different temperature. After 25 minutes, he measured the percentage of light passing through the water in each tube.

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The student put the same volume of water in each tube.

Explain why it was important that he controlled this experimental variable.

[2 marks]

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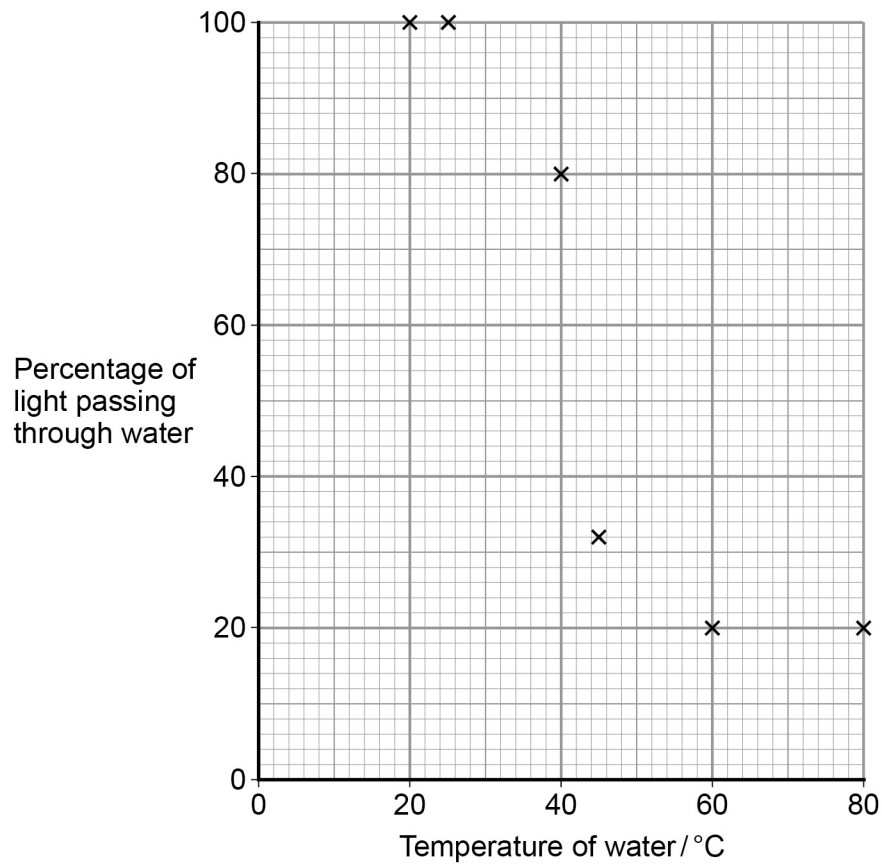
Describe a method the student could have used to monitor the temperature of the water in each tube.

[1 mark]



Figure 2 shows the student's results.

Figure 2



0 2 . 3

Draw a suitable curve on **Figure 2**.

[1 mark]

0 2 . 4

The decrease in the percentage of light passing through the water between 25 °C and 60 °C is caused by the release of the red pigment from cells of the beetroot.

Suggest how the increase in temperature of the water caused the release of the red pigment.

[2 marks]

